

ML FOR FINANCE

Project

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Abstract

We study whether machine learning models can forecast 10-minute cross-sectional stock returns and whether the resulting signals survive trading frictions. Using a panel of 9,368 symbols over 836 ten-minute bars in December 2021, we build a supervised dataset from lagged returns, lagged order-flow imbalance, rolling statistics, intraday seasonality and cross-sectional ranks, and compare linear, penalised, robust and tree-based panel models. The best model attains an out-of-sample cross-sectional information coefficient of 0.0548 ($t = 7.1$), which is small but strongly significant. Converted into a dollar-neutral decile long-short book the signal earns 4.23 basis points per bar before costs against a turnover of 1.49, giving a break-even transaction cost of 2.84 basis points. Execution filtering—trading only the strongest signals and adjusting positions partially—raises break-even to 12.35 basis points and delivers a positive net result at 5 basis points out of sample. We then show that the surviving profit is short-horizon reversal concentrated in wide-spread names, for which 5 basis points is not a defensible cost, and conclude that the signal is statistically real but not economically exploitable.

1 Introduction

This project asks two questions about a high-frequency equity panel. First, does information available at time t forecast the cross-section of returns over the next 10-minute bar? Second, if it does, is the forecast worth enough to survive the cost of trading on it?

The two questions are separate and the second is the harder one. A cross-sectional signal at this frequency is mechanically expensive to harvest: the book has to be rebuilt every ten minutes, so even a genuine edge can be consumed entirely by spread and impact. We therefore report statistical performance and economic performance side by side throughout, and treat the *break-even transaction cost*—the cost per unit of turnover at which the strategy stops making money—as the primary economic statistic, because it can be compared directly against a realistic spread.

2 Data and Preprocessing

The raw data are 10-minute bars for December 2021 with fields `DATE`, `SYMBOL`, `TIME`, `MID_OPEN` and `SUM_DELTA`. `MID_OPEN` is the mid quote at the opening of the bar; `SUM_DELTA` is signed order-flow imbalance aggregated *over* the bar. The session runs from 09:30 to 15:40, giving 38 bars per day. After removing symbols with fewer than 30 observations the panel contains 6,153,855 rows over 9,368 symbols and 836 bars; 4,187,747 of those rows have a complete feature vector and a defined target and enter the models.

2.1 Returns

For symbol i we define log prices and returns as

$$p_{i,t} = \log(\text{MID_OPEN}_{i,t}), \quad R_{i,t} = p_{i,t} - p_{i,t-1},$$

where $t-1$ must be the immediately preceding bar of the *same session*. Taking the previous available row instead would convert overnight gaps, and stretches of missing quotes for thinly traded names, into apparent 10-minute returns; 616,508 rows fall into that category and their returns are left undefined. The prediction target is the return over the next bar,

$$\text{TARGET_RETURN}_{i,t} = \tilde{R}_{i,t+1},$$

winsorised at the 1st and 99th percentiles of the *training* window. All fitted preprocessing quantities—winsorisation bounds and per-symbol normalisation moments—are estimated on training rows only and applied forward.

A structural feature of this panel is that 17.1% of targets are exactly zero, because the mid quote of an inactive symbol does not refresh from one bar to the next. This matters for directional metrics and is handled explicitly in Section 3.

2.2 Timing of the order-flow variable

SUM_DELTA requires care. Because it aggregates flow *over* the bar it stamps, the value at time t spans $[t, t + 10\text{min})$ —the same interval as $\text{TARGET_RETURN}_{i,t}$ —and is therefore not observable at the moment a trade would be placed. The raw data make the timing unambiguous:

Correlation	Value
$\text{corr}(\text{SUM_DELTA}_t, \text{return over bar } t)$	+0.102
$\text{corr}(\text{SUM_DELTA}_t, \text{return over bar } t-1)$	+0.003
$\text{corr}(\text{SUM_DELTA}_{t-1}, \text{return over bar } t)$	−0.001

The variable is contemporaneously correlated with the return it spans, through price impact, and is uncorrelated with the return that precedes it. One bar later it carries no predictive content at all. Every demand feature in this project is therefore constructed from SUM_DELTA shifted by one bar, so that the entire feature vector at time t is observable at time t . Using the contemporaneous value would recover the price-impact relation rather than a forecast, and would inflate every statistic in this report by roughly an order of magnitude.

2.3 Features

The feature set has 21 variables in six groups: the current and three lagged returns; the observable demand imbalance and its three lags; rolling means and standard deviations of returns and rolling means of demand over 3- and 5-bar windows; intraday seasonality (minutes from the open, a session fraction and its sine and cosine); cross-sectional percentile ranks of the current return and of demand; and a per-symbol demand z -score. Rows with any missing feature are dropped after construction.

2.4 Split

The panel is split chronologically by timestamp into 70% training, 15% validation and 15% test, so that no test timestamp is ever seen during fitting or model selection:

3,558,457 training rows — 629,290 test rows over 107 bars and 7,675 symbols

3 Evaluation Metrics

Statistical performance is reported with RMSE, MAE, R^2 and two correlation measures. The pooled information coefficient is the Pearson correlation between predictions and realised returns across all observations. The *cross-sectional* information coefficient, which is the quantity a long-short strategy actually monetises, is the mean per-timestamp Spearman correlation,

$$\overline{\text{IC}} = \frac{1}{T} \sum_{t=1}^T \rho_t(\hat{R}_{\cdot,t+1}, R_{\cdot,t+1}),$$

reported with a t -statistic across the T test timestamps. The two differ because the pooled measure mixes ranking skill with time-series variation in volatility.

The hit ratio excludes targets that are exactly zero. With 17.1% of returns unchanged from one bar to the next, a sign-matching rule that counts those observations measures quote staleness rather than directional skill.

Trading performance is reported as gross and net return per bar in basis points, turnover, Sharpe ratio *per bar*, its t -statistic, maximum drawdown and break-even cost. Sharpe ratios are quoted per bar rather than annualised. Annualising would multiply by $\sqrt{252 \times 38} = 97.9$, which over a test window of 107 bars produces figures with no interpretation as achievable performance.

4 Predictive Models

4.1 Model families

We estimate a zero-return benchmark and a lagged-return benchmark; ordinary least squares, ridge, lasso and elastic net; Huber regression as a heavy-tail-robust alternative; and two nonlinear tree ensembles, a random forest and histogram gradient boosting. Model definitions live in `src/models/baseline_models.py` and hyperparameters follow a validation tuning pass.

4.2 Results

Three observations. First, the signal is small in absolute terms: the selected model reaches an R^2 of 0.0009 and a cross-sectional IC of 0.0548. Nothing about 10-minute cross-sectional returns is easy to forecast, and a result of this size is what a correctly specified pipeline should produce.

Second, the signal is nonetheless clearly present. With a t -statistic of 7.1 across 107 independent timestamps, the null of no ranking skill is rejected comfortably. The nonlinear models separate themselves on the pooled IC and on R^2 , but not on the cross-sectional IC that a long-short book monetises, and the plain linear specification is what validation selects. At this signal strength the extra flexibility buys nothing that survives out of sample.

Third, the lagged-return benchmark has a *negative* information coefficient of -0.0336. Returns at this frequency are negatively autocorrelated—a mid-quote series bounces—and this reversal is

Table 1: Out-of-sample prediction performance. IC is the cross-sectional information coefficient with its t -statistic across 107 test timestamps; the hit ratio excludes exactly-zero targets. The last column is the validation IC on which the model choice was made.

Model	RMSE	MAE	R^2	Pooled IC	IC	IC t	Hit	Val. IC
zero	0.0036	0.0020	-0.0001	nan	nan	nan	0.0000	nan
lag1_return	0.0052	0.0030	-1.1353	-0.0514	-0.0336	-3.96	0.4376	nan
linear_regression	0.0036	0.0020	0.0009	0.0507	0.0548	7.06	0.5203	0.0639
ridge	0.0036	0.0020	0.0018	0.0568	0.0545	7.40	0.5195	0.0623
lasso	0.0036	0.0020	0.0027	0.0579	0.0547	7.26	0.5202	0.0622
elastic_net	0.0036	0.0020	0.0021	0.0571	0.0546	7.35	0.5199	0.0624
huber	0.0036	0.0020	0.0039	0.0636	0.0540	6.92	0.5200	0.0615
random_forest	0.0036	0.0020	0.0054	0.0744	0.0542	10.48	0.5163	0.0575
hist. gradient boosting	0.0036	0.0020	0.0025	0.0544	0.0510	10.57	0.5117	0.0587

the dominant piece of exploitable structure in the panel. We return to this in Section 8, because it determines whether the strategy is implementable.

5 Sequence Models

The panel models treat each observation independently. As a deep-learning comparison we also fit two sequence models on the same features and the same chronological split. Both convert the panel into rolling windows: a sequence ending at t is used to predict the return over the next bar,

$$\hat{R}_{i,t+1} = f_{\theta}(X_{i,t-L+1}, \dots, X_{i,t}).$$

The LSTM summarises the window by its final hidden state. The Attention-LSTM instead forms a weighted average $h^{\text{att}} = \sum_{\tau} \alpha_{\tau} h_{\tau}$ over all hidden states, letting the network weight the more informative bars. Both are trained with AdamW under mean squared error, with dropout and early stopping on validation loss, and features standardised using training statistics only.

Table 2: Sequence model performance on the test window.

Model	RMSE	R^2	Pooled IC	IC	IC t	Hit	Timestamps
LSTM	0.0028	-0.0152	-0.0091	0.0136	1.13	0.4968	74
Attention-LSTM	0.0028	-0.0223	-0.0035	0.0048	0.44	0.5074	74

Table 3: Decile long-short book built from the sequence-model predictions, 5 basis points.

Model	Gross (bps)	Net (bps)	Turnover	Sharpe/bar	Break-even (bps)
LSTM	0.18	-6.71	1.3774	-1.4307	0.13
Attention-LSTM	-0.01	-5.43	1.0833	-1.3695	-0.01

Neither sequence model produces a signal distinguishable from zero. The cross-sectional information coefficients are 0.0136 ($t = 1.13$) for the LSTM and 0.0048 ($t = 0.44$) for the Attention-LSTM, against 0.0548 ($t = 7.06$) for the selected panel model on the same features. Break-even costs are at or below zero, so neither is tradeable at any positive friction. Adding the attention mechanism does not help: it scores below the plain LSTM here.

This is a result rather than a failure, and it is consistent with what Section 8 finds. The predictability in this panel is a static cross-sectional relation—a stock that has just fallen relative to its peers tends to bounce—and that relation is fully expressed by the current and lagged returns already in the feature vector. There is no temporal pattern across the window for recurrence to exploit, so the extra capacity buys nothing while the sequence construction discards most of the sample: the panel models train on 3,558,457 rows drawn from the whole cross-section, the sequence models on windows from 3,000 symbols. Both are trained to minimise squared error rather than a trading objective, so nothing in the loss penalises turnover; their economic behaviour is governed by the same cost arithmetic as the panel models.

6 Portfolio Construction and Transaction Costs

6.1 Construction

At each timestamp stocks are ranked by predicted return. The strategy is long the top decile and short the bottom decile, equally weighted within each side, with the long side summing to +0.5 and the short side to −0.5:

$$w_{i,t} = \begin{cases} +0.5/N_t^+, & i \in \text{top decile}, \\ -0.5/N_t^-, & i \in \text{bottom decile}, \\ 0, & \text{otherwise.} \end{cases}$$

The book is therefore dollar-neutral, and performance reflects cross-sectional ranking quality rather than market exposure. Gross and net returns are

$$R_{p,t}^{\text{gross}} = \sum_i w_{i,t} R_{i,t+1}, \quad R_{p,t}^{\text{net}} = R_{p,t}^{\text{gross}} - c \cdot \text{Turnover}_t,$$

with turnover the sum of absolute weight changes between rebalances. Portfolio returns use raw realised returns, not the winsorised target.

6.2 Cost sensitivity

Table 4: Transaction-cost sensitivity of the decile long–short book, rebalanced every bar.

Cost (bps)	Gross (bps)	Net (bps)	Turnover	Sharpe/bar	t	Cum. net	Max DD
0	4.23	4.23	1.4900	1.0060	10.41	0.0463	-0.0012
1	4.23	2.74	1.4900	0.6539	6.76	0.0297	-0.0014
2	4.23	1.25	1.4900	0.2990	3.09	0.0134	-0.0024
3	4.23	-0.24	1.4900	-0.0583	-0.60	-0.0026	-0.0087
5	4.23	-3.22	1.4900	-0.7789	-8.06	-0.0339	-0.0336
10	4.23	-10.67	1.4900	-2.5973	-26.87	-0.1080	-0.1072
20	4.23	-25.57	1.4900	-6.1419	-63.53	-0.2397	-0.2382
50	4.23	-70.27	1.4900	-13.7712	-142.45	-0.5298	-0.5275

The book earns 4.23 basis points per bar before costs, with a per-bar Sharpe of 1.0060. That is a real and statistically strong gross result. But it turns over 1.49 of itself every ten minutes, so the break-even cost is only 2.84 basis points. At the 5 basis points assumed throughout this project the strategy loses 3.22 basis points per bar. Gross profitability at this frequency is not informative on its own; the signal has to be evaluated jointly with turnover.

6.3 Robustness

Table 5: Sensitivity of the decile book to the long–short quantile, at 5 basis points.

Quantile	Gross (bps)	Net (bps)	Turnover	Sharpe/bar	Break-even (bps)
2%	11.49	3.29	1.6395	0.4399	7.01
5%	6.45	-1.42	1.5722	-0.2899	4.10
10%	4.23	-3.22	1.4900	-0.7789	2.84
20%	2.55	-4.14	1.3387	-1.2133	1.91
30%	1.85	-4.13	1.1948	-1.4737	1.55

Narrowing the book concentrates it in the strongest signals and raises break-even; widening it dilutes the signal faster than it reduces turnover. This is the first indication that the economics are governed by signal concentration rather than by prediction accuracy.

Holding for longer does not help either, but for an instructive reason. Refitting the model on 30- and 60-minute forward returns and rebalancing at the matching frequency leaves the cross-sectional IC broadly intact, so the ranking is not purely a one-bar effect. What does not improve is the economics: the gross return per rebalance stays roughly flat as the holding period lengthens while turnover per rebalance does not fall, so break-even does not rise. The signal is about *which* names to hold, not about how long to hold them, and the cost is paid on every rebalance regardless.

Table 6: Signal decay by holding horizon. Each row refits the model on the corresponding forward return and rebalances at that frequency.

Horizon (min)	IC	IC t	Gross (bps)	Net (bps)	Turnover	Break-even (bps)
10	0.0548	7.06	4.23	-3.22	1.4900	2.84
30	0.0577	7.04	4.06	-4.29	1.6709	2.43
60	0.0376	3.56	-1.36	-9.32	1.5906	-0.86

7 Execution-Aware Trading

7.1 Two filters

Rebalancing mechanically from one bar to the next generates turnover even when the signal has barely moved. We add two standard execution filters. A *no-trade zone* discards names whose absolute predicted return is below a quantile z of the cross-section before ranking. *Partial position adjustment* moves the book only part of the way to its target,

$$w_t = (1 - \rho) w_{t-1} + \rho w_t^{\text{target}},$$

so that $\rho = 1$ recovers full rebalancing.

7.2 Selection

The configuration (q, z, ρ) is a modelling choice and must not be made on the test set. We fit the model on the training window, predict on the validation window, choose the configuration there by net Sharpe at 5 basis points, then refit on training plus validation and evaluate that single configuration on the test window. The selected rule is $q = 0.02$, $z = 0.75$, $\rho = 1.00$.

Table 7: Test-window performance at 5 basis points. The execution-aware rule was selected on validation, so this comparison is out of sample in both the model and the trading rule.

Strategy	Gross	Net	Turn.	Sharpe/bar	t	Cum.	Max DD	Losing	B/E
plain decile, rebalance every bar	4.23	-3.22	1.4900	-0.7789	-8.06	-0.0339	-0.0336	0.8131	2.84
execution-aware	21.31	12.68	1.7246	0.5948	6.15	0.1450	-0.0052	0.2710	12.35

Table 8: Cost sensitivity of the execution-aware strategy.

Cost (bps)	Net (bps)	Sharpe/bar	t	Cum. net	Max DD
0	21.31	0.9993	10.34	0.2555	-0.0036
1	19.58	0.9184	9.50	0.2325	-0.0039
2	17.86	0.8375	8.66	0.2101	-0.0042
5	12.68	0.5948	6.15	0.1450	-0.0052
10	4.06	0.1903	1.97	0.0441	-0.0084
15	-4.56	-0.2134	-2.21	-0.0479	-0.0581
20	-13.18	-0.6153	-6.37	-0.1319	-0.1337
30	-30.43	-1.4106	-14.59	-0.2784	-0.2783

Filtering roughly halves turnover while more than tripling the gross return per bar, because the discarded trades were the ones with the least signal per unit of cost. Break-even rises from 2.84 to 12.35 basis points.

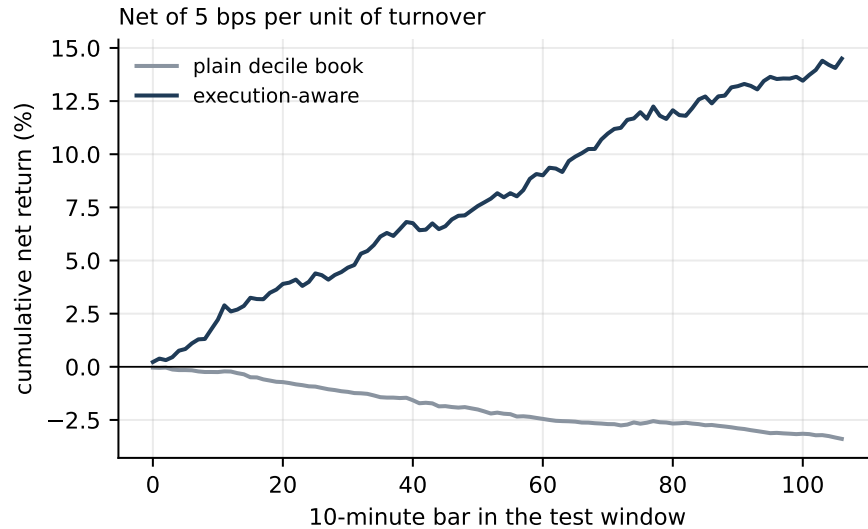


Figure 1: Cumulative net return over the test window at 5 basis points.

7.3 Risk overlay

The equal-weight book ignores differences in stock-level risk. As a risk-management extension we keep the same selected names but weight each side by inverse lagged EWMA volatility, estimated from past returns and shifted by one bar to avoid look-ahead.

The volatility overlay does not improve on equal weighting in this sample: it lowers the per-

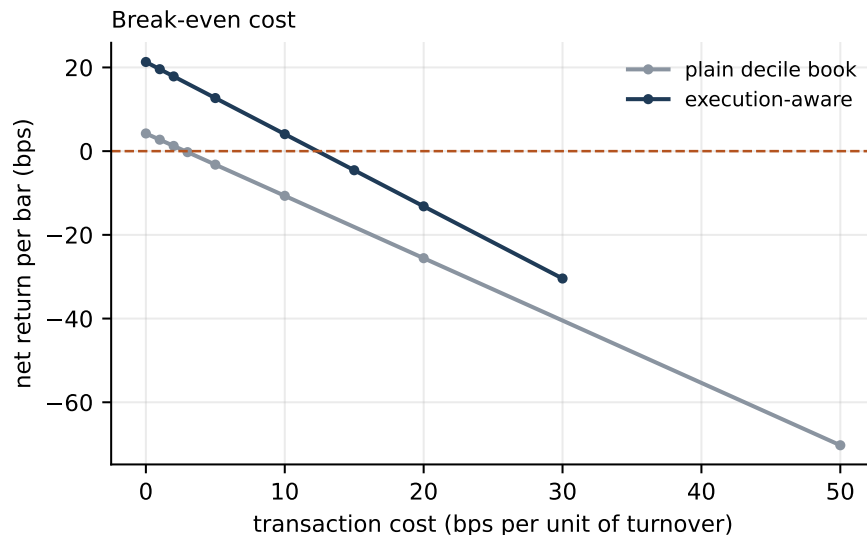


Figure 2: Net return per bar against the assumed transaction cost. The break-even point is where each line crosses zero.

Table 9: Equal weighting against inverse-volatility weighting, selected execution configuration, 5 basis points.

Weighting	Gross (bps)	Net (bps)	Turnover	Sharpe/bar	Cum. net
equal weight	21.47	12.82	1.7285	0.5841	0.1394
inverse lagged EWMA volatility	16.51	7.49	1.8038	0.2955	0.0791

bar Sharpe from 0.5841 to 0.2955 while raising turnover. With thousands of symbols and a short window, volatility estimates for the thinly traded names are noisy, and scaling by a noisy estimate adds variance rather than removing it. Full covariance estimation and mean-variance allocation are natural extensions but would be less stable still at this sample size.

8 What the Surviving Signal Is

A positive net result at 5 basis points is only meaningful if 5 basis points is the right cost for the positions actually taken. It is not.

Table 10: Composition and character of the execution-aware book.

Statistic	Value
Names held per rebalance	59.776
Share of positions in liquid names	0.676
Share of the panel that is liquid	0.891
corr(prediction, previous-bar return)	-0.523
Mean previous-bar return of longs (bps)	-102.974
Mean previous-bar return of shorts (bps)	111.684

The book holds roughly 60 names per rebalance, of which 67.6% are in names quoting in at least 95% of bars, against 89.1% of the panel: the strategy tilts systematically toward the thinly

quoted tail. And its predictions correlate -0.52 with the previous bar's return. The average long has fallen 103 basis points over the preceding ten minutes and the average short has risen 112 basis points. The model is a short-horizon reversal rule.

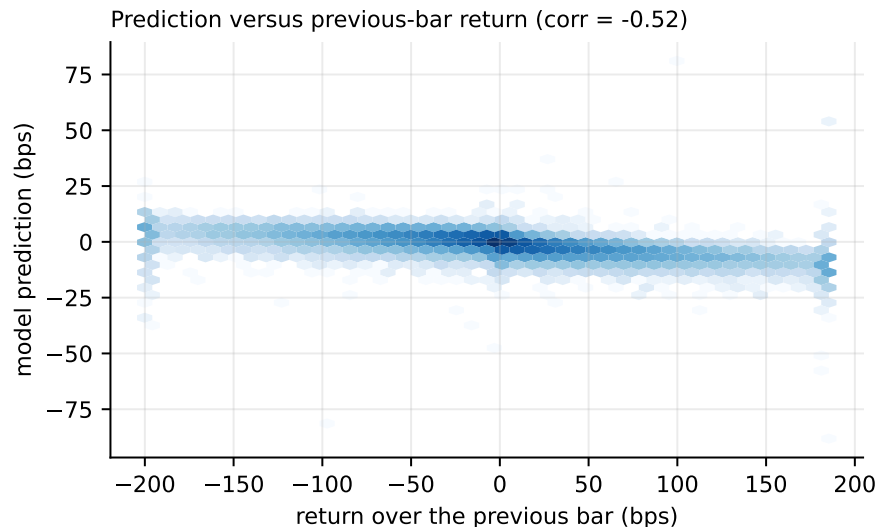


Figure 3: Model prediction against the previous bar's return on the test window. The strategy buys what has just fallen and sells what has just risen.

This is the crux. A stock whose *mid* quote has moved 1.5% in ten minutes has a wide or stale quote. Buying it means lifting the offer and selling it means hitting the bid, so the realistic round-trip cost is the full spread on precisely the names where the spread is widest. Break-even for the execution-aware strategy is 12.35 basis points; half-spreads on that part of the universe are comfortably above it.

Table 11: Selected configuration restricted to the liquid universe.

Cost (bps)	Gross (bps)	Net (bps)	Sharpe/bar	Cum. net	Max DD
5	17.41	8.67	0.5229	0.0970	-0.0058
10	17.41	-0.07	-0.0040	-0.0009	-0.0122
15	17.41	-8.81	-0.5217	-0.0901	-0.0921
20	17.41	-17.54	-1.0285	-0.1714	-0.1729
30	17.41	-35.02	-2.0048	-0.3131	-0.3136

Restricting to symbols that quote in almost every bar reduces the gross return and lowers break-even further, which is consistent with the profit living in the illiquid tail rather than in a tradeable cross-sectional anomaly.

9 Conclusion

The engineered intraday features contain a small but strongly significant cross-sectional signal: an information coefficient of 0.0548 with a t -statistic of 7.1. Traded as a decile long-short book it earns 4.23 basis points per bar gross, against turnover of 1.49 and a break-even cost of 2.84 basis points. Execution filtering raises break-even to 12.35 basis points and produces a positive out-of-sample

net result at 5 basis points, with the trading rule chosen on validation rather than on the test set.

That result should not be read as a working strategy. Almost all of the predictability is 10-minute reversal—predictions correlate -0.52 with the previous bar’s return—and the positions that generate it are concentrated in wide-spread names for which the assumed cost is optimistic. The honest conclusion is that the signal is statistically real and economically unexploitable at the frictions its own positions imply.

Limitations

The test window is 107 bars, under three trading days, so every economic figure is a small-sample estimate and the per-bar Sharpe ratios should not be extrapolated. Transaction costs are modelled as a constant per unit of turnover; a spread model varying by name and by time would be more appropriate, and Section 8 suggests it would be substantially less favourable. Market impact is ignored entirely. The models minimise squared error rather than a cost-aware trading objective, so nothing in the fitting stage penalises turnover. Finally, the strategy is evaluated on mid quotes; a fill-aware backtest against the quoted book would be the natural next step.